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Local Crowding-Out in China

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1 Big picture

- **Question.** Does local public debt crowd out private investment in China, and if so, through which financing channel?
- **Setting.** The paper studies Chinese prefecture-level cities between 2006 and 2013, combining a newly constructed city-level database on local government debt with firm-level manufacturing data.
- **Core challenge.** A negative correlation between local debt and private investment is not enough: local governments may borrow more in bad times, or common local shocks may raise both public borrowing and private investment. The paper therefore needs within-city sources of heterogeneity that isolate financing frictions from broader local conditions.
- **Main result.** Local public debt crowds out the investment of private manufacturing firms, while leaving state-owned firms largely unaffected.
- **Main mechanism.** The crowding-out operates through *tighter local credit constraints*. Banks absorb local government debt and reduce credit to riskier, less politically connected borrowers—especially private and small firms.
- **Why China is a useful setting.** China’s credit market is geographically segmented. Local banks operate within narrow geographic boundaries, interest-rate controls limit price adjustment, and local public debt is largely absorbed by local banks. As a result, public borrowing can crowd out *local* private credit rather than being smoothed nationally.
- **Three identification strategies.** The paper uses:
 - within-city firm location relative to banks in neighboring cities,
 - industry-level external-finance dependence in the spirit of Rajan and Zingales,
 - cash-flow sensitivity of investment, including an endogenous switching-regression split between constrained and unconstrained firms.
- **Main empirical findings.** Private firms invest less in cities with higher local public debt; the effect is weaker for firms near banks in nearby cities, stronger for industries that need more external finance, stronger for small firms, and stronger in cities where the return to capital is high. State-owned firms are largely spared.
- **Main quantitative fact.** Local government debt rose from about 5.8% of GDP in 2006 to about 22% in 2013. The paper’s baseline city-level and firm-level regressions imply economically meaningful declines in private investment, and the cash-flow-sensitivity estimates show that credit constraints tighten sharply as debt rises.
- **Why this paper matters.** It links China’s post-2008 fiscal stimulus to local credit reallocation and to capital misallocation. The implication is not only less private investment, but a shift of credit away from more productive private firms toward the public sector and politically connected firms.

2 Data and measurement (key objects)

2.1 City-level local government debt panel

- **Unit of observation.** Chinese prefecture-level cities, which are large administrative units containing an urban core plus surrounding districts.
- **Coverage.** The authors collect debt data for all 293 prefecture-level cities from 2006 to 2013. The estimation sample contains 261 cities because macroeconomic controls are unavailable for 32 cities.
- **What is measured as local debt.** The paper measures the total liabilities of local government financing vehicles (LGFVs), including both bank loans and bonds.
- **Why LGFVs matter.** Chinese municipalities cannot normally borrow directly. Instead, they set up LGFVs, transfer assets such as land to them, and have these vehicles borrow from banks or issue bonds.
- **Construction of the debt data.** The authors exploit the fact that LGFVs issuing bonds must disclose balance sheets for the current year and at least the prior three years. This allows them to back out total debt outstanding, not just bond issuance.
- **Validation.** When aggregated to the national or provincial level, the constructed debt series closely matches official National Audit Office figures, though it is best interpreted as a conservative lower bound.
- **Key magnitude.** Local government debt rose from roughly 1.2 trillion yuan in 2006 to 12.5 trillion yuan in 2013, and the debt-to-GDP ratio increased from about 5.8% to about 22%.

2.2 Other city-level variables

- **Macroeconomic controls.** The paper uses city-level GDP, GDP growth, GDP per capita, population, budget balance, total bank loans, and land prices.
- **Financial-depth proxy.** A key variable is

$$BL_{ct} = \frac{\text{bank loans in city } c \text{ at } t}{\text{GDP in city } c \text{ at } t}.$$

It proxies for local financial depth and is used both directly and in interactions with cash flow or industry dependence.

- **Return to capital.** The paper also uses a city-level measure of the return to capital, denoted RC , to study whether crowding-out is more severe where capital markets are especially segmented.

2.3 Firm-level panel and variables

- **Main firm data.** The core plant-level source is the Annual Survey of Industrial Firms (ASIF), also known as the Chinese Industrial Enterprise Database.
- **Coverage.** ASIF covers manufacturing firms above a sales threshold: 5 million yuan until 2009 and 20 million yuan thereafter.

- **Additional source.** Because ASIF is unavailable in 2010, the paper merges in the Annual Tax Survey (ATS) to recover additional firm financial information for 2007–2011.
- **Final sample.** After cleaning and winsorization, the final sample contains about 1.16 million firm-year observations on more than 300,000 firms.
- **Core firm-level variables.** Investment I , cash flow CF , revenue growth REV , leverage LEV , total assets, age, ownership, and Altman-style distance to default (Z -score).
- **Investment measure.** Fixed investment is scaled by beginning-of-year total assets.
- **Ownership split.** The paper distinguishes **private** firms from **state-owned** firms because state firms enjoy preferential access to bank credit and implicit public guarantees.

2.4 Within-city geography and nearby-bank measures

- **Border proximity.** BD_i is a dummy for firms close to the border of another city.
- **Nearby-bank access.** BK_i is a dummy equal to one if the average distance between firm i and the 10 closest bank branches located in another city is below a threshold. The paper also uses a continuous proximity measure PX_i .
- **Interpretation.** These variables capture whether a firm can potentially tap credit outside its own city and therefore escape local crowding-out.
- **Neighbor-city controls.** The paper constructs neighboring-city debt, lending, and growth variables ($NLGD$, NBL , NGR) to make sure the results are not driven by conditions in the nearby city itself.

2.5 Credit-market segmentation as an identifying background fact

- **Local banks are local.** City and rural financial institutions rarely operate outside their own city or province, which mechanically segments credit markets.
- **Large-bank internal capital markets are limited.** Even large banks face lending quotas, loan-to-deposit constraints, and frictions in reallocating funds across branches.
- **Interest-rate regulation matters.** Because loan and deposit rates are regulated, local public borrowing does not trigger a local interest-rate response that would otherwise draw in outside funds.
- **Empirical evidence.** The paper shows that the cross-city dispersion in the return to capital remains large and that even local government bond yields retain city-specific residual variation after controlling for maturity, issue size, ratings, and time effects.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline city-level correlation between investment and local debt

The paper begins with the city-level regression

$$I_{ct} = \beta LGD_{ct} + X'_{ct}\gamma + \alpha_c + \tau_t + \varepsilon_{ct}, \quad (1)$$

where I_{ct} is the city-level manufacturing investment ratio, LGD_{ct} is local government debt over GDP, and X_{ct} contains city controls such as bank loans, budget balance, GDP growth, GDP per capita, population, and land prices.

- **Interpretation.** This is the most reduced-form test: do cities with more public debt have lower manufacturing investment?
- **What it captures.** A negative β is consistent with local crowding-out.
- **Limitation.** This specification is vulnerable to endogeneity from reverse causality and common local shocks.

3.2 Firm-level investment regressions

The firm-level version is

$$I_{ict} = \beta LGD_{ct} + X'_{ict}\gamma + \alpha_i + \xi_c + \tau_t + u_{ict}, \quad (2)$$

where I_{ict} is firm i 's investment ratio, X_{ict} includes lagged investment, revenue growth, and lagged cash flow, and the fixed effects control for firm, city, and year heterogeneity.

- **Interpretation.** This moves from city aggregates to within-firm variation and allows the authors to compare private vs. state and large vs. small firms.
- **Why it helps.** Firm fixed effects absorb time-invariant differences in technology, management, and firm type.
- **Key empirical use.** The paper augments this regression with $STATE$ and $STATE \times LGD$ to test whether the negative relation is weaker for state-owned firms.

3.3 Why simple OLS is not enough

The authors emphasize two endogeneity concerns. In words, the true system may look like

$$I = \alpha + \beta D + \varepsilon, \quad D = a + bI + u,$$

where investment may itself affect debt issuance and the two equations may share common shocks.

- **Reverse causality.** If local governments borrow more when private investment is weak, the estimated crowding-out coefficient can be biased.

- **Common shocks.** Public infrastructure booms or local growth shocks can increase both debt and private demand conditions.
- **Implication.** The sign of the OLS bias is ambiguous, which is exactly why the paper relies on within-city heterogeneity and saturated fixed effects in later sections.

3.4 Within-city location design: access to outside-city banks

To exploit within-city geographic heterogeneity, the paper estimates

$$I_{ict} = (\delta_1 BK_i + \delta_2 BD_i) \times LGD_{ct} + X'_{ict}\gamma + \alpha_i + \theta_{ct} + u_{ict}, \quad (4)$$

where θ_{ct} are city-year fixed effects.

- **Interpretation.** The main effect of local debt is fully absorbed by city-year fixed effects. Identification comes only from whether firms in the *same city-year* but with different access to neighboring-city banks respond differently.
- **Economic logic.** A positive δ_1 means that firms closer to banks in other cities are less hurt by local debt issuance.
- **Why this is powerful.** It rules out many omitted-variable concerns because all city-year shocks are absorbed.
- **Key refinement.** The paper shows that proximity to *nearby-city banks* matters more than simple proximity to the city border, suggesting a financing mechanism rather than general access to other inputs.

3.5 Industry external-finance dependence design

At the city-industry-year level, the paper estimates

$$I_{jct} = \beta I_{jc,t-1} + \delta(EF_j \times LGD_{ct}) + \alpha_{jt} + \theta_{ct} + \eta_{cj} + u_{jct}, \quad (5)$$

where EF_j is an industry-level measure of external-finance dependence.

- **Interpretation.** The coefficient δ measures whether local debt hurts investment more in industries that technologically need more external funding.
- **Identification strategy.** The inclusion of city-year, industry-year, and city-industry fixed effects means the estimate comes from differential responses across industries within the same city and year.
- **Why this matters.** This is the Rajan-Zingales logic adapted to local debt crowding-out. If debt tightens credit supply, the effect should be concentrated in externally dependent industries.

3.6 Cash-flow sensitivity of investment

The key cash-flow-sensitivity regression is

$$I_{ict} = \beta I_{i,c,t-1} + \alpha REV_{i,c,t-1} + (\gamma_1 + \gamma_2 LGD_{ct}) CF_{i,c,t-1} + \alpha_i + \theta_{ct} + u_{ict}. \quad (6)$$

- **Interpretation of γ_1 .** This is the baseline sensitivity of investment to internal funds.
- **Interpretation of γ_2 .** A positive γ_2 means that as local public debt rises, firms' investment becomes more dependent on cash flow—that is, financing constraints tighten.
- **Why city-year fixed effects matter.** Since θ_{ct} absorb the direct effect of local debt and all other city-year shocks, the estimate comes from within-city-year differences in firms' ability to rely on internal finance.
- **Extension.** The paper also adds $CF \times BL$ to show that greater local bank depth reduces cash-flow sensitivity, consistent with the interpretation that BL proxies for local credit availability.

3.7 Cash-flow sensitivity and the return to capital

The paper also studies whether crowding-out is especially severe in segmented places by interacting cash flow and debt with the city-level return to capital:

$$CF_{i,t-1} \times LGD_{ct} \times RC_{ct}.$$

- **Interpretation.** If RC is high, capital is especially scarce in that city. A positive coefficient on the triple interaction means that local debt is most harmful exactly where local capital scarcity is already severe.
- **Economic meaning.** This links the credit-crowding channel to the efficiency cost of misallocation: the adverse effect is larger where the shadow value of capital is high.

3.8 Switching regression with endogenous sample separation

To avoid relying only on exogenous splits such as private/state or small/large, the paper estimates a two-regime switching model:

$$W_{ict}^* = Z_{ict}'\mu + u_{ict}, \quad (7)$$

$$I_{ict}^{(1)} = X_{ict}'\alpha_1 + \varepsilon_{1,ict}, \quad (8)$$

$$I_{ict}^{(2)} = X_{ict}'\alpha_2 + \varepsilon_{2,ict}. \quad (9)$$

Here W_{ict}^* is a latent index governing whether a firm is in the unconstrained regime (1) or the constrained regime (2), and Z_{ict} includes age, size, distance to default, tangibility, private ownership, and local government debt.

- **Interpretation.** The model jointly estimates which firms are likely to be constrained and how their investment responds to cash flow and local debt.
- **Why it matters.** This avoids imposing an arbitrary sample split and lets the data determine which firms behave as constrained borrowers.
- **Prediction.** If the paper's mechanism is right, the positive interaction between cash flow and local debt should appear in the constrained regime, not in the unconstrained regime.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A city with weak private investment may induce local governments to borrow more for counter-cyclical reasons.
- Conversely, a city with strong public-investment opportunities may experience both higher debt and local private spillovers.
- Therefore, the raw city-level correlation between debt and investment is not by itself causal evidence of crowding-out.

4.2 Why the Chinese institutional setting helps identification

- Local public debt is mostly funded by local banks through LGFVs.
- Credit markets are segmented geographically, especially because local banks operate in narrow territories.
- Interest-rate ceilings prevent a local debt shock from being equilibrated through local rate increases that would attract outside funds.
- As a result, local public borrowing plausibly creates a localized credit squeeze rather than a purely national financing event.

4.3 Within-city spatial heterogeneity

- The location design compares firms within the same city-year, holding city-level shocks fixed.
- The key identifying assumption is that conditional on the controls and city-year effects, proximity to banks in neighboring cities only matters because it gives firms alternative financing access.
- The fact that bank proximity dominates border proximity strengthens this interpretation: the effect seems to be about finance rather than border-adjacent access to labor, land, or other inputs.

4.4 Industry external-finance dependence

- The Rajan-Zingales-style strategy uses technological variation across industries.
- Because the regression includes city-year and industry-year fixed effects, the estimate does not rely on comparing different cities with different macro shocks or different industries with different national trends.
- It asks a sharper question: within the same city and year, do externally dependent industries suffer more when local debt is high?

4.5 Cash-flow sensitivity and endogenous regime selection

- The exogenous sample-split approach compares private vs. state and small vs. large firms.
- The endogenous switching model goes one step further and estimates the probability that a firm is constrained using age, size, tangibility, default risk, ownership, and local debt.
- The fact that local debt raises cash-flow sensitivity for constrained firms but not for unconstrained firms is strong evidence for the financing-constraint channel.

4.6 Relation to the literature

- Relative to the debt-and-growth literature, the paper identifies a local causal mechanism rather than a broad cross-country correlation.
- Relative to Rajan and Zingales, it uses local public debt as the source of credit tightening and local Chinese data to construct the external-finance index.
- Relative to Love (2003), it adapts the cash-flow-sensitivity framework to a setting where local public debt itself is the source of changing financing constraints.
- Relative to the China misallocation literature, it offers direct evidence that the post-2008 stimulus reallocated credit away from private firms toward the public sector and state-connected borrowers.

5 Tables and figures

5.1 Figure 1 and Table I: the new debt data and its magnitude

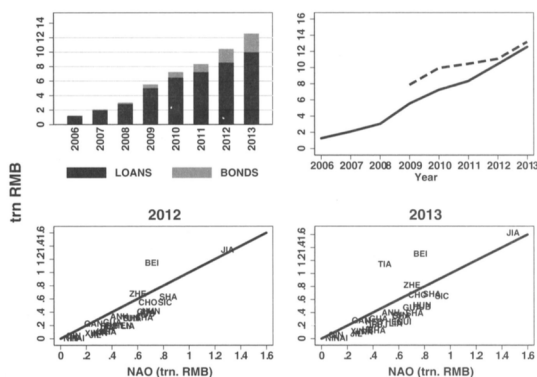


Figure 1. Local government debt in China. Top left panel: composition of local government debt. Top right panel: total local government debt according to our data (solid line) and NAO data (dashed line). Bottom panels: our data for province-level local government debt plotted against NAO data in 2012 (left) and 2013 (right).

**Table I
Local Government Debt in China**

This table summarizes our data for local government debt. Columns (2) to (5) are based on city-level variables. Columns (6) and (7) report annual totals in RMB and as a % of China's GDP.

Year	μ	σ	Min.	Max.	Total China		N. Cities	
	Bill. RMB				Bill. RMB	(% GDP)	All	$D > 0$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
2006	4.3	18.1	0.0	173	1,255	5.7	293	92
2007	7.1	27.6	0.0	268	2,087	7.9	293	144
2008	10.4	38.4	0.0	383	3,036	9.7	293	189
2009	18.9	62.8	0.0	589	5,535	15.8	293	248
2010	24.7	80.5	0.0	789	7,249	17.4	293	281
2011	28.5	93.7	0.0	951	8,336	16.8	293	291
2012	35.6	113.0	0.0	1,145	10,425	18.8	293	292
2013	42.9	132.1	0.0	1,303	12,556	20.1	293	291

Read:

- The debt database is a major contribution of the paper. Figure 1 shows that the authors' series tracks official NAO debt figures closely at both the national and provincial levels.
- The top-left panel of Figure 1 also shows that local debt is mostly **bank loans**, not bonds. This matters because bank-funded local debt is exactly the kind most likely to crowd out local private credit.

- Table I documents the speed of the debt surge: local debt rose from about 1.2 trillion yuan in 2006 to 7.2 trillion in 2010 and 12.5 trillion in 2013.
- The key message is that the post-2008 local public debt boom was not a marginal financing episode—it was massive and geographically widespread.

5.2 Figure 2: geographic segmentation of China's credit market

Read:

- The top panel shows that the dispersion in the return to capital across cities remains large and even rises around 2009, despite reforms that might have been expected to integrate capital markets.
- The bottom panel shows that residual yields on LGFV bonds are systematically related to local credit-market depth: local governments in shallower credit markets pay more even after controlling for bond characteristics and ratings.
- This figure is the paper's institutional backbone. It makes the localized crowding-out story plausible.

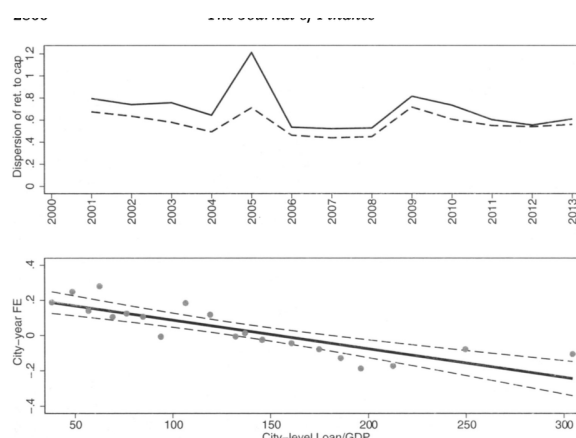


Figure 2. Geographic segmentation of China's credit market. Top panel: time series of the between-cities coefficient of variation in the return to capital; the solid line plots the raw data, and the dashed line plots the data after a 5% Winsorization of the return to capital. Bottom panel: binned scatterplot of estimated city-time effects from a regression of the residuals of LGFV bond yield residuals against the loan to GDP ratio of the corresponding cities and years.

Table III
Investment and Local Government Debt: City-Level Regressions

This table reports results of regressions in which the dependent variable is the city-level investment ratio of the manufacturing sector (computed as the weighted average of investment scaled by total assets of all manufacturing firms in a given city and year) and the dependent variables are local government debt over GDP (*LGD*), bank loans scaled by GDP (*BL*), local government balance scaled by GDP (*GB*), GDP growth (*GR*), the log of GDP per capita (*GDP PC*), the log of population (*POP*), and the log of the price of land (*LP*). Columns (1) and (2) include all manufacturing firms, column (3) private-sector manufacturing firms only, column (4) state-owned manufacturing firms only, column (5) large manufacturing firms only, and column (6) small manufacturing firms only. Robust standard errors clustered at the firm and city-year levels are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>LGD</i>	-0.083** (0.033)	-0.093* (0.041)	-0.104** (0.039)	-0.029 (0.050)	-0.081* (0.040)	-0.229** (0.080)
<i>BL</i>		-0.011 (0.022)	-0.002 (0.025)	-0.027 (0.016)	-0.008 (0.021)	-0.028 (0.074)
<i>GB</i>		0.019 (0.218)	0.028 (0.234)	-0.139 (0.242)	0.055 (0.233)	-0.430 (0.600)
<i>GR</i>		0.408** (0.146)	0.332* (0.143)	0.632*** (0.163)	0.424** (0.140)	0.084 (0.310)
<i>ln(GDP PC)</i>		4.858* (2.542)	6.761* (3.228)	-5.851* (3.060)	2.919 (2.153)	18.121 (11.760)
<i>ln(POP)</i>		7.889** (3.077)	9.774* (3.822)	-5.674 (3.237)	5.761* (2.480)	27.243* (13.491)
<i>ln(LP)</i>		0.583 (0.561)	0.489 (0.564)	-0.411 (0.992)	0.568 (0.603)	1.624 (2.406)
N. Obs.	1,862	1,800	1,798	1,514	1,800	1,800
N. Cities	261	261	261	261	261	261
Firms	All	All	Private	State	Large	Small
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes

5.3 Table III: city-level investment and debt

Read:

- City-level investment is negatively correlated with local debt. In the baseline specification, the coefficient on *LGD* is around -0.09 for all firms and -0.10 for private firms.
- The coefficient is small and statistically insignificant for state-owned firms.
- The effect is much stronger for small firms: the coefficient is about -0.23 , nearly three times the full-sample magnitude.

Table IV
Capital Productivity and Local Government Debt

This table reports results of regressions in which the dependent variable is the city-level capital productivity of the manufacturing sector (computed as the average percentage deviation of firm-level capital productivity from the industry mean) and the dependent variables are local government debt scaled by GDP (*LGD*), bank loans scaled by GDP (*BL*), local government balance scaled by GDP (*GB*), GDP growth (*GR*), the log of GDP per capita (*GDP PC*), the log of population (*POP*), and the log of the price of land (*LP*). Columns (1) and (2) report estimates obtained using the sample of all manufacturing firms, columns (3) and (4) report estimates based on the subsample of private-sector manufacturing firms, and columns (5) and (6) report estimates based on the subsample of state-owned manufacturing firms. Robust standard errors clustered at the firm and city-year levels are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>LGD</i>	0.236*** (0.087)	0.271*** (0.091)	0.251*** (0.090)	0.275*** (0.094)	0.099 (0.172)	0.075 (0.194)
<i>BL</i>		0.026 (0.059)		0.019 (0.062)		0.178* (0.091)
<i>GB</i>		-1.206** (0.579)		-0.914 (0.571)		-1.923* (1.011)
<i>GR</i>		0.209 (0.319)		0.256 (0.331)		-0.096 (0.855)
$\ln(\text{GDP PC})$		17.891* (10.293)		14.673 (10.533)		49.286** (21.094)
$\ln(\text{POP})$		52.470*** (17.591)		53.205*** (18.136)		64.286** (28.770)
$\ln(\text{LP})$		0.352 (2.124)		0.609 (2.187)		-4.573 (3.961)
N. Obs.	782	739	782	739	782	739
N. Cities	260	257	260	257	260	257
Firms	All	All	Private	Private	State	State
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes

- The message is already clear here: the negative relation is concentrated in firms that are most plausibly credit constrained.

5.4 Table IV: capital productivity and debt

Read:

- Capital productivity is *higher* in cities with more local debt, especially for private firms.
- This is exactly what one expects if public borrowing crowds out private investment and leaves productive private firms with too little capital.
- The absence of a comparable effect for state-owned firms reinforces the idea that the mechanism is about differential access to credit.

5.5 Table V and Table VI: firm-level investment and leverage

Read:

- Table V moves to firm-level data and again finds a negative relationship between local debt and investment: in the baseline regression, the *LGD* coefficient is about -0.055 .
- The interaction $STATE \times LGD$ is positive, meaning the negative debt-investment relation is significantly weaker for state-owned firms.
- Table VI shows the financing side directly: leverage is negatively related to local debt for private firms but not for state-owned firms, and leverage is also negatively related to the local share of bank lending going to LGFVs.
- The combination of Tables V and VI says the result is not just lower investment demand—it is reduced access to credit.

Table V

Investment and Local Government Debt: Firm-Level Regressions

This table reports results of regressions in which the dependent variable is the firm-level investment ratio (computed as investment scaled by total assets at the beginning of the year), and the explanatory variables are the lagged investment ratio (I_{t-1}), change in revenue scaled by total assets (REV_{t-1}), lagged cash flow scaled by total assets (CF_{t-1}), a state ownership dummy ($STATE$), local government debt scaled by city-level GDP (LGD), and the interaction between LGD and $STATE$. The regressions in columns (1)-(3) control for firm, city, and year fixed effects, while the regression in column (4) controls for firm and city-year fixed effects. Robust standard errors clustered at the firm and city-year levels are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
I_{t-1}	-0.271*** (0.006)	-0.271*** (0.006)	-0.271*** (0.006)	-0.274*** (0.002)
REV_{t-1}	4.050*** (0.089)	4.050*** (0.089)	4.050*** (0.089)	3.772*** (0.079)
CF_{t-1}	7.779*** (0.519)	7.780*** (0.519)	7.780*** (0.519)	6.987*** (0.195)
$STATE$		-0.396** (0.174)	-0.697*** (0.224)	-0.253 (0.176)
LGD	-0.055*** (0.016)	-0.055*** (0.016)	-0.056*** (0.016)	
$STATE \times LGD$			0.027*** (0.009)	0.013* (0.007)
N. Obs.	1,035,427	1,035,427	1,035,427	1,035,400
N. Cities	261	261	261	261
Firm FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	No
Year FE	Yes	Yes	Yes	No
City-Year FE	No	No	No	Yes
$LGD + STATE \times LGD$			-0.029	
p-Value			0.12	

Table VI

Firm Leverage, Local Government Debt, and Share of Local Bank Lending to LGFVs

This table reports results of regressions in which the dependent variable is firm-level leverage, and the explanatory variables are local government debt scaled by GDP (LGD), local bank lending to local government financing vehicles scaled by total local bank lending to corporates ($LGFV$), total bank loans scaled by GDP (BL), budget balance (GB), log of GDP per capita ($\ln(GDP\ PC)$), GDP growth (GR), land price (LP), and firm size ($SIZE$). Columns (1), (4), (7), and (8) report estimates based on the sample of all manufacturing firms, columns (2) and (5) focus on the subsample of private manufacturing firms, and columns (3) and (6) focus on the subsample of state-owned manufacturing firms. The specifications of columns (7) and (8) also control for city-level return to capital (RC). Robust standard errors clustered at the firm and city-year levels are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LGD	-0.009** (0.004)	-0.013*** (0.004)	-0.001 (0.015)					
$LGFV$				-0.073*** (0.006)	-0.077*** (0.007)	0.003 (0.027)	-0.066*** (0.022)	
BL	0.025*** (0.001)	0.029** (0.002)	-0.006 (0.007)	0.022*** (0.002)	0.025*** (0.002)	-0.006 (0.007)	0.024*** (0.006)	0.027*** (0.006)
GB	-0.067*** (0.020)	-0.063*** (0.022)	-0.234*** (0.069)	-0.077*** (0.020)	-0.075*** (0.022)	-0.235*** (0.069)	-0.058 (0.073)	-0.046 (0.075)
$\ln(GDP\ PC)$	-2.610*** (0.214)	-2.776*** (0.238)	-0.278*** (0.821)	-2.671*** (0.214)	-2.789*** (0.238)	-0.277 (0.821)	-4.682*** (1.050)	-4.710*** (1.053)
GR	0.058 (0.011)	0.065*** (0.013)	-0.121*** (0.044)	0.068*** (0.011)	0.072*** (0.013)	-0.121*** (0.044)	0.102*** (0.036)	0.089** (0.038)
LP	0.163*** (0.060)	0.057 (0.070)	0.735*** (0.230)	0.111* (0.060)	-0.006 (0.070)	0.735*** (0.230)	0.216 (0.170)	0.248 (0.175)
$SIZE$	-0.454*** (0.050)	-1.245*** (0.057)	-1.677*** (0.264)	-0.446*** (0.050)	-1.228*** (0.057)	-1.677*** (0.264)	-0.561* (0.335)	-0.568* (0.336)
RC							0.932 (2.389)	0.619 (2.455)
N. Obs.	751,974	591,084	40,332	751,974	591,084	40,332	591,152	591,152
N. Cities	261	261	261	261	261	261	261	261
Sample	All	Private	State	All	Private	State	All	All
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

5.6 Table VII: within-city proximity to outside-city banks

Read:

- The interaction $LGD \times BK$ is positive and significant. Firms close to banks in neighboring cities are less exposed to local crowding-out.
- Border proximity also matters, but once both are included together, proximity to nearby-city banks dominates.
- The interpretation is important: crowding-out seems to operate through **finance**, not through general access to nonfinancial inputs across borders.
- The effect survives the use of a continuous proximity measure and controls for neighboring-city debt, lending, and growth.

5.7 Table VIII and Figure 3: external-finance dependence

Read:

- The interaction between external-finance dependence and local debt is negative and statistically significant.
- The result is concentrated in private firms; it is small and insignificant for state-owned firms.
- Figure 3 translates the coefficient into economically intuitive terms: for a low-dependence industry such as paper, investment is barely affected by debt; for a high-dependence industry such as batteries, investment falls markedly as local debt rises.
- In the paper's calibration from the regression, investment in the high-dependence industry drops

Table VII
Investment, Local Government Debt, and Proximity to Other Cities

This table reports results of regressions in which the dependent variable is the firm-level investment ratio, and the explanatory variables are the lagged investment ratio (I_{t-1}), change in revenue scaled by total assets (REV_{t-1}), lagged cash flow scaled by total assets (CF_{t-1}), a dummy variable equal to 1 only for firms for which the average distance of the 10 closest bank branches located in another city is less than 20 km (BK), a dummy variable equal to 1 for firms located less than 20 km from the city border (BD), the interaction between each of BK and BD and local government debt scaled by GDP (LGD), bank loans over GDP (BL), and GDP growth (GR). In column (4), BK is replaced with a continuous measure of proximity (PX) defined as 100 minus the average distance of the 10 closest banks located in another city. Columns (6) and (7) also control for the interaction between BK and government debt ($NLGD$), growth (NGR), and bank loans (NBL) in the city in which the neighboring banks are located. Robust standard errors clustered at the firm, city-year, and neighboring city-year (in columns (7) and (8)) levels are reported in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
I_{t-1}	-0.258*** (0.006)	-0.258*** (0.006)	-0.258*** (0.006)	-0.258*** (0.006)	-0.259*** (0.006)	-0.263** (0.007)	-0.263** (0.007)
REV_{t-1}	2.187*** (0.047)	2.187*** (0.047)	2.187*** (0.047)	2.187*** (0.047)	2.186*** (0.048)	2.210*** (0.054)	2.224*** (0.056)
CF_{t-1}	4.409*** (0.330)	4.408*** (0.330)	4.412*** (0.330)	4.408*** (0.330)	4.437*** (0.333)	4.650*** (0.371)	4.642*** (0.378)
$LGD \times BK$	0.017* (0.008)	0.022** (0.008)			0.028** (0.012)	0.031** (0.012)	0.036** (0.015)
$LGD \times BD$	0.015 (0.010)		0.023** (0.010)				
$LGD \times PX$				0.004*** (0.001)			
$GR \times BK$					0.023 (0.021)		0.087** (0.044)
$BL \times BK$					-0.003 (0.003)		-0.007 (0.004)
$NLGD \times BK$						-0.002 (0.014)	-0.007 (0.017)
$NGR \times BK$							-0.089* (0.047)
$NBL \times BK$							0.006* (0.004)
N. Obs.	792,900	792,900	792,900	792,900	769,328	603,127	581,973
N. Cities	251	251	251	251	251	251	251
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
C-Y FE	No	Yes	Yes	Yes	Yes	Yes	Yes
N. C-Y FE	No	No	No	No	No	Yes	Yes

Table VIII
Industry-Level Regressions

This table reports results of regressions in which the dependent variable is the investment ratio (computed as investment over total assets at the beginning of the year) aggregated at the city-industry-year level. The regressions control for the initial investment ratio (I_{t-1}) and the interaction between the Rajan-Zingales index of external financial dependence (EF) computed on firms in Beijing, Shanghai, Hangzhou, and Wenzhou and each of the following variables: local government debt over GDP (LGD), bank loans over GDP (BL), the log of GDP per capita ($GDP\ PC$), GDP growth (GR), and the log of average land price (LP). The regression in column (1) reports estimates based on the sample of all manufacturing firms and the EF index computed using Chinese data, while column (2) reports estimates based on the same sample and on the original Rajan-Zingales EF index. The regressions reported in the subsequent columns are all based on the EF index computed using Chinese data but are estimated on different subsamples: private-sector manufacturing firms in column (3), state-owned manufacturing firms in column (4), firms in industries with below-median exposure to government expenditure in column (5), and firms in industries with above-median exposure to government expenditure in column (6). Robust standard errors clustered at the city-industry level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
I_{t-1}	-0.216*** (0.007)	-0.168*** (0.011)	-0.213*** (0.007)	-0.394*** (0.031)	-0.232*** (0.011)	-0.221*** (0.011)
$EF \times LGD$	-0.017*** (0.008)	-0.014** (0.007)	-0.017** (0.001)	-0.004 (0.015)	-0.018** (0.009)	-0.011 (0.012)
N. Obs.	46,379	18,398	44,527	3,655	21,461	17,370
N. Cities	257	257	257	197	256	256
With Additional Interactions						
I_{t-1}	-0.217*** (0.006)	-0.174*** (0.011)	-0.214*** (0.007)	-0.398*** (0.111)	-0.234*** (0.001)	0.220*** (0.011)
$EF \times LGD$	-0.021*** (0.007)	-0.017*** (0.006)	-0.021*** (0.007)	-0.007 (0.079)	-0.023*** (0.009)	-0.012 (0.011)
$EF \times BL$	0.004*** (0.001)	0.017*** (0.001)	0.004*** (0.001)	0.001 (0.006)	0.004 (0.002)	0.006*** (0.002)
$EF \times \ln(GDP\ PC)$	0.4078* (0.22)	-0.543*** (0.166)	0.352 (0.223)	0.788 (2.501)	0.456 (0.327)	-0.062 (0.380)
$EF \times GR$	0.025 (0.019)	0.104*** (0.013)	0.030 (0.020)	0.063 (0.189)	0.067* (0.034)	-0.015 (0.034)
$EF \times LP$	-0.174 (0.112)	0.408*** (0.106)	-0.175 (0.121)	-0.311 (1.353)	-0.018 (0.180)	-0.213 (0.187)
N. Obs.	45,753	18,138	43,958	3,554	21,161	17,138
N. Cities	257	257	257	197	255	255
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-City FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	All	All	Private	State	Low Exp.	High Exp.

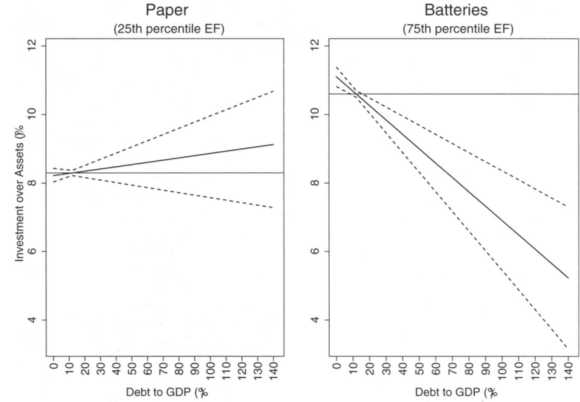


Figure 3. Local government debt and investment ratios in different industries. The figure shows how investment ratios vary with the level of government debt for manufacturing firms in the paper industry (25th percentile of the distribution of the index of external financial dependence) and the battery industry (75th percentile of the distribution of the index of external financial dependence). The graphs are based on the estimations of Table VIII, column (3). The dashed lines are 95% confidence intervals and the horizontal lines are the average investment ratios of the two industries (8.3% for paper and 10.6% for batteries).

below its average once debt exceeds roughly 15% of GDP and falls to about 9% when debt reaches 50% of GDP.

5.8 Table IX, Table X, and Figure 4: cash-flow sensitivity

Table IX
Cash-Flow Sensitivity of Investment

This table reports results of regressions in which the dependent variable is the firm-level investment ratio (computed as investment scaled by total assets at the beginning of the year), and the explanatory variables are the lagged investment ratio (I_{t-1}), the change in revenue scaled by total assets (REV_{t-1}), lagged cash flow scaled by total assets (CF_{t-1}), and the interaction between CF_{t-1} and local government debt scaled by GDP (LGD). Column (1) reports estimates based on the full sample of manufacturing firms, column (2) those based on the subsample of private-sector manufacturing firms, column (3) those based on the subsample of state-owned manufacturing firms, column (4) those based on the subsample of large firms (top 25% of the distribution by assets), and column (5) those based on the subsample of small firms (bottom 25% of the distribution by assets). Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)
I_{t-1}	-0.273*** (0.002)	-0.280*** (0.002)	-0.371*** (0.008)	-0.230*** (0.004)	-0.333*** (0.005)
REV_{t-1}	3.773*** (0.031)	3.799*** (0.034)	2.398*** (0.167)	5.955*** (0.117)	1.954*** (0.057)
CF_{t-1}	6.725*** (0.231)	7.334*** (0.256)	4.328*** (1.190)	5.815*** (0.660)	4.472*** (0.539)
$CF_{t-1} \times LGD$	0.028** (0.011)	0.029** (0.013)	-0.097 (0.055)	-0.020 (0.026)	0.075* (0.030)
N. Obs.	1,035,400	858,624	45,922	110,091	107,694
N. Cities	261	261	261	261	261
Firm FE	Yes	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes	Yes
Sample	All	Private	State	Large	Small

Table X
Cash-Flow Sensitivity of Investment: Controlling for Bank Loans

This table reports results of regressions in which the dependent variable is the firm-level investment ratio (computed as investment over total assets at the beginning of the year), and the explanatory variables are the lagged investment ratio (I_{t-1}), the change in revenue scaled by total assets (REV_{t-1}), lagged cash flow scaled by total assets (CF_{t-1}), and the interaction between CF_{t-1} and each of the following variables: local government debt scaled by GDP (LGD) and bank loans scaled by GDP (BL). Column (1) reports estimates based on the full sample of manufacturing firms, column (2) those based on the subsample of private-sector manufacturing firms, column (3) those based on the subsample of state-owned manufacturing firms, column (4) those based on the subsample of large firms (top 25% of the distribution by assets), and column (5) those based on the subsample of small firms (bottom 25% of the distribution by assets). In the specification reported in column (6), cash flow is interacted with a dummy (LB) that equals 1 if in the relevant city the share of the branches of the four largest Chinese banks in the total number of city branches exceeds the sample median, and 0 otherwise. Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
I_{t-1}	-0.274*** (0.002)	-0.281*** (0.002)	-0.371*** (0.008)	-0.230*** (0.004)	-0.333*** (0.005)	-0.274*** (0.002)
REV_{t-1}	3.770*** (0.031)	3.796*** (0.033)	2.393*** (0.168)	5.954*** (0.135)	1.951*** (0.067)	3.774*** (0.031)
CF_{t-1}	8.343*** (0.374)	9.141*** (0.411)	6.020*** (1.902)	6.367*** (1.193)	7.062*** (1.037)	10.073*** (0.447)
$CF_{t-1} \times LGD$	0.075*** (0.014)	0.082*** (0.016)	-0.044 (0.069)	-0.016 (0.038)	0.157*** (0.045)	0.073*** (0.017)
$CF_{t-1} \times BL$	-0.022** (0.004)	-0.025*** (0.004)	-0.023 (0.019)	-0.007 (0.011)	-0.035*** (0.011)	-0.031** (0.004)
$CF_{t-1} \times LGD \times LB$						-0.033 (0.028)
$CF_{t-1} \times BL \times LB$						0.034*** (0.007)
$CF_{t-1} \times LB$						-5.258*** (0.708)
N. Obs.	1,035,400	868,624	45,922	110,091	107,694	1,035,383
N. Cities	261	261	261	261	261	261
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	All	Private	State	Large	Small	All

Read:

- In Table IX, the interaction $CF \times LGD$ is positive for the full sample and for private firms, but insignificant for large firms and opposite-signed for state-owned firms.
- Figure 4 is especially revealing: for state-owned firms, the cash-flow sensitivity of investment falls with local debt and becomes statistically indistinguishable from zero when debt reaches around 20% of GDP. They effectively become unconstrained.
- For small firms, the opposite happens: the cash-flow sensitivity rises sharply with local debt, consistent with tightened financing constraints.
- Table X adds bank-loan interactions and strengthens the point. Once local bank depth is controlled for, the $CF \times LGD$ coefficient in the full sample rises from about 0.028 to about 0.075, while $CF \times BL$ is negative. This is strong evidence that debt tightens constraints and local bank depth relaxes them.

5.9 Table XI and Table XII: where the squeeze is strongest and who is constrained

Read:

- Table XI shows that the cash-flow-sensitivity effect is much stronger in cities with above-median return to capital. In high-return cities, the $CF \times LGD$ coefficient is large and positive; in

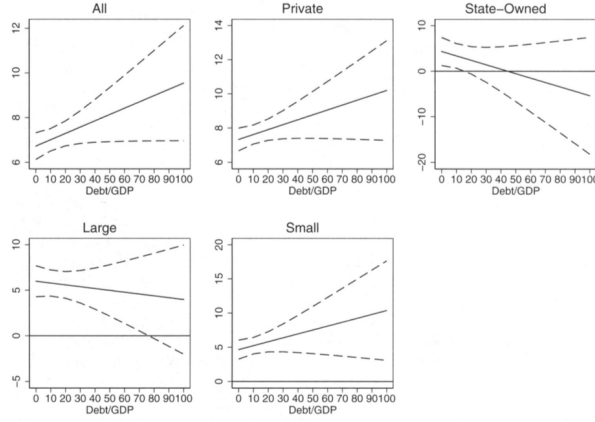


Figure 4. Investment sensitivity to cash flow. The figure shows how the sensitivity of investment to cash flow changes with the level of local government debt. These marginal effects are based on the estimates reported in Table IX.

Table XI
Cash-Flow Sensitivity of Investment and the Return to Capital

This table reports results of regressions in which the dependent variable is the firm-level investment ratio (computed as investment scaled by total assets at the beginning of the year), and the explanatory variables are the lagged investment ratio (I_{t-1}), change in revenue scaled by total assets (REV_{t-1}), lagged cash flow scaled by total assets (CF_{t-1}), and the interaction between CF_{t-1} and local government debt scaled by GDP (LGD). Columns (1) and (2) report estimates for the subsamples of manufacturing firms located in cities with above- and below-median return to capital, respectively. Columns (3) and (4) report estimates for the same models of columns (1) and (2) for the subsample of private-sector firms. Column (5) interacts government debt and cash flow with a continuous measure of city-level return to capital (RC), and column (6) also adds interactions with city-level bank loans scaled by GDP (BL). Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)
I_{t-1}	-0.304*** (0.002)	-0.228*** (0.004)	-0.313*** (0.003)	-0.235*** (0.004)	-0.272*** (0.002)	-0.272*** (0.002)
REV_{t-1}	3.611*** (0.048)	4.151*** (0.068)	3.655*** (0.053)	4.137*** (0.072)	3.782*** (0.036)	3.782*** (0.036)
CF_{t-1}	7.597*** (0.330)	6.500*** (0.435)	8.254*** (0.377)	6.851*** (0.472)	6.907*** (0.238)	6.791*** (0.238)
$CF_{t-1} \times LGD$	0.167*** (0.033)	-0.018 (0.020)	0.162*** (0.037)	-0.019 (0.022)	0.051** (0.021)	0.048*** (0.021)
$CF_{t-1} \times RC$					-29.889*** (3.065)	-32.054*** (3.170)
$CF_{t-1} \times LGD \times RC$					0.896*** (0.228)	0.868*** (0.228)
$CF_{t-1} \times BL$						3.002*** (0.6667)
$CF_{t-1} \times BL \times RC$						25.651*** (8.521)
N. Obs	469,038	219,659	373,025	188,808	764,769	764,769
N. Cities	147	143	147	143	171	171
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	High Ret. Cities All Firms	Low Ret. Cities All Firms	High Ret. Cities Private	Low Ret. Cities Private	All Cities All Firms	All Cities All Firms

Table XII
Switching Regression Model

This table reports results of the switching regression model described in equations (7) to (9). The selection equation (Panel A) controls for the log of firm age ($\ln(Age)$), the log assets ($\ln(Assets)$), distance to default ($Zscore$), a time-invariant industry-level measure of the share of tangible assets over total assets ($Tangible$), a dummy that takes a value of 1 for private-sector firms ($Private$), and a time-variant measure of city-level local government debt (LGD). The coefficients (and standard errors) in the selection equation are multiplied by 100 to facilitate readability. The investment equation (Panel B) controls for lagged cash flow (CF), the interaction between lagged cash flow and local government debt (LGD), lagged investment (not reported), and revenue growth (not reported). Model 1 includes city and year fixed effects, Model 2 includes city-year fixed effects, and Model 3 includes city-year and industry-year fixed effects. For each model, we report separate investment equations for firms that are not credit-constrained (regime 1) and credit-constrained firms (regime 2). Robust standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A. Selection Equation						
	(1)	(2)	(3)			
$\ln(Age)$	10.770*** (0.077)	7.176*** (0.072)	8.437*** (0.066)			
$\ln(Assets)$	0.396** (0.034)	0.685*** (0.002)	1.680*** (0.027)			
$Zscore$	0.097*** (0.018)	0.994*** (0.016)	0.918*** (0.011)			
$Private$	-8.943*** (0.141)	-5.063*** (0.132)	-4.248*** (0.117)			
$Tangible$	8.401*** (0.280)	4.642*** (0.261)				
LGD	-0.012* (0.001)					
N. Obs	1,060,404	1,060,404	1,060,404			
Panel B. Investment Equation						
	(1.1) Not Constr.	(1.2) Constr.	(2.1) Not Constr.	(2.2) Constr.	(3.1) Not Constr.	(3.2) Constr.
CF_{t-1}	15.65*** (0.030)	4.158*** (0.183)	2.969*** (0.240)	8.208*** (0.187)	13.931*** (3.231)	7.125*** (0.028)
$CF_{t-1} \times LGD$	-0.410*** (0.001)	0.142*** (0.009)	-0.056*** (0.001)	0.047*** (0.010)	-0.333*** (0.01)	0.114*** (0.001)
LGD	-0.013*** (0.001)	-0.041*** (0.004)				
N. Obs.	305,603	754,800	2745,048	785,355	231,967	828,436
City FE	Yes		No		No	
Year FE	Yes		No		No	
City-Year FE	No		Yes		Yes	
Ind-Year FE	No		No		Yes	

low-return cities, it is near zero.

- This means crowding-out is most severe where capital is already scarce and therefore where the efficiency cost is greatest.
- Table XII's switching regression confirms the mechanism from a different angle. In the constrained regime, investment is highly cash-flow sensitive and this sensitivity rises with local debt. In the unconstrained regime, the interaction is negative.
- The selection equation also lines up with intuition: unconstrained firms are older, larger, safer, more tangible, and less likely to be private.

6 Short summary

- **Conclusion.** The paper shows that local public debt in China crowded out private manufacturing investment between 2006 and 2013.
- **Identification insight.** The causal mechanism is pinned down using within-city firm heterogeneity: access to nearby-city banks, industry external-finance dependence, and cash-flow sensitivity.
- **Economic lesson.** The crowding-out works by tightening local credit constraints, and it falls mainly on private and small firms, while state-owned firms are largely protected.
- **Broader implication.** The fiscal stimulus increased aggregate investment in the short run, but it also reallocated credit toward the public sector and politically connected firms, thereby worsening capital allocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Local public debt in China does not merely raise aggregate borrowing; it changes who gets financed.
- Because Chinese credit markets are geographically segmented, local public borrowing is absorbed by local banks and crowds out local private borrowers.
- The empirical evidence is consistent across many designs: private firms invest less, borrow less, and become more cash-flow dependent when local debt is high, while state-owned firms do not.

7.2 Economic intuition

- When local governments issue more debt through LGFVs, local banks devote more balance-sheet capacity to the public sector.
- Banks then ration credit toward the borrowers who are riskier, less collateralized, or less politically connected—in China, typically private and small firms.

- Firms that can escape the local market by borrowing near another city's banks are less affected, which is why the spatial design is so informative.
- The result is a reallocation of capital away from higher-productivity private firms and toward the public sector and state-affiliated firms.

7.3 Takeaway

This paper is best read as a clean local-credit-market version of the broader China misallocation story. Its message is not simply that government debt can crowd out private investment in the abstract. The paper shows *how* that crowding-out happens: through segmented local bank balance sheets, under interest-rate and institutional frictions, and with effects concentrated on the private firms that are most likely to be productive but financially constrained. The broader implication is that fiscal stimulus financed through local public debt can support aggregate activity in the short run while simultaneously weakening long-run growth by worsening the allocation of credit.